Curriculum vitæ

Personal information

Mattia Capuano
Place and date of birth: Garbagnate Milanese (Italy), September 13th, 1999
Nationality: Italy
Email address: m.capuano@herts.ac.uk

Education

- 10/2018-12/2021: **Bachelor's degree in Physics**
  University of Milan
  Final grade: 110/110 cum laude
  Thesis: *Quantum field theory in curved spacetime: the case of Inflation*
  Supervisor: prof. Emanuele Castorina

- 12/2021-22/09/2023: **Master's degree in Physics**
  University of Milan
  Final grade: 110/110 cum laude
  Thesis: *On the nature of soft and collinear scales in Higgs boson production at NNLO*
  Supervisor: prof. Stefano Forte

- 10/2024- *ongoing*: **PhD Maths**
  University of Hertfordshire
  Research area: *Positive and negative geometries for scattering amplitudes*
  Principal Supervisor: dr. Tomasz Łukowski

Papers and preprints

- *Canonical Differential Equations for Cosmology from Positive Geometries,* MC, L. Ferro, T. Łukowski, A. Palazio, arXiv:2505.14609;
- *Cosmology meets cluster algebra,* MC, L. Ferro, T. Łukowski, A. Palazio, arXiv:2512.14859m, published in JHEP 11 (2026) 010;
- *Generalised cluster adjacency for cosmology,* MC, L. Ferro, T. Łukowski, A. Palazio, arXiv:2603.09965;
- Book chapter *The Disordered Universe 2025: Analytic Properties of Amplitudes and Correlators,* edited by Springer, with A. McLeod et al. (https://arxiv.org/pdf/2610.36271).

Recently attended Conferences, Meetings and Workshops

- Feb. 2024 – Galaxies meet QCD, Zurich;
- Dec. 2024 – Milan Christmas meeting 2024, Milan;
- June 2025 – FPUK meeting 2025, Edinburgh;
- Dec. 2025 – Symbology@15, Munich;
- Dec. 2025 – Milan Christmas meeting 2025, Milan;
- May 2026 – Prague Spring Amplitudes 2026, Prague;
- June 2026 – Amplitudes 2026, London;
- July 2026 – MiaPPbP workshop, *Primordial Cosmology: Novel Perspectives from Scattering Amplitudes, Holography and the Bootstrap*, Munich;
- September 2026 – Pollica Physics Workshop *Scattering amplitudes: from positive geometries to analytic structures,* Pollica (Italy);
- November 2026 – Loop-the-Loop 2026, Pisa.

Recently attended Schools

- March 2025 - CAPP 2025 (Computer Algebra and Particle Physics), Hamburg;
- July 2025 - *Summer School The Disordered Universe 2025* - Castro Caldelas, Spain.
- June 2026 – Nordfjordeid Summer School 2026, *Positive Geometry and Scattering Amplitudes,* Norway;
- August 2026 – KMPB School 2026, *Symbols et Periods*, Berlin;

Talks and Posters

- July 2025 - FPUK meeting 2025, poster: *Canonical Differential Equations for Cosmology from Positive Geometries*;
- Sep. 2025 - MTP Research Day 2025, talk: *Positive geometries and scattering amplitudes*;
- Dec. 2025 - Milan Christmas meeting 2025: *Differential equations for cosmological correlators*;
- Aug. 2026 - KMPB School 2026, poster: *Generalised cluster adjacency for Cosmology*;
- Sep. 2026 – Pollica Physics Workshop, talk: *Cosmology meets Cluster Algebra;*
- Nov. 2026 – Loop-the-Loop, talk.

Teaching

- 2024/25 to 2025/26 – *Nonlinear systems* (BSc Maths), tutorials;
- 2025/26 – *Mathematical methods* (BSc Maths), tutorials;
- 2024/25 to 2025/26 – *Maths Support Service*;
- 2024/25 to 2026/27 – Seminars for the MSc students at the University of Hertfordshire: *Canonical quantization in curved spacetime, Is it always possible to parametrize a curve?, Differential equations for cosmological correlators, Iterated integrals and multiple polylogs, Feynman Integrals: from Integration-by-Parts to Twisted Cohomology.*

Organisation

- 2025/26 to 2026/27 - Organiser of the Research Seminars Module, University of Hertfordshire.
- Organiser of *CASE 2026: Correlators and Amplitudes in the South East*. One-day meeting funded by GradNet.

Awarded Scholarships

- 2021/22 - Regional university scholarship, amount: 2007.00 EUR;
- 2022/23 - Regional university scholarship, amount: 3427.00 EUR;
- 2022/23 - Graduating award, amount: 1713.50 EUR;
- 2024/25 - Herts' Funding to Support Postgraduate Research, amount: 750 GBP;
- 2025/26 - Herts' Funding to Support Postgraduate Research, amount: 750 GBP.